

ORBITAL AND PHOTOMETRIC PROPERTIES OF SZ LYNCIS

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ABSTRACT

We report new photometric results based on 3760 observations made on the Johnson *BVRI* system and 69 new radial-velocity measurements of the large-amplitude δ Scuti binary SZ Lyncis. Using these data and previous observations from the literature, improved values for the pulsation and orbital parameters are derived. We find the pulsation period to be 0.12052115 days, which is shorter than earlier determinations. The pulsation period is undergoing a secular period change of 3×10^{-12} days cycle $^{-1}$. The orbital period of the binary is found to be 1181.1 days, with an orbital eccentricity of 0.188. A new ephemeris is given, which accurately predicts the times of light maximum and is consistent with the spectroscopic orbit. The mass function is $0.101 M_{\odot}$, which, together with other known properties of SZ Lyn, indicates that the unseen companion is most likely on the main sequence with a spectral type between F2 and K3. The orbital inclination is probably greater than 47° . All of our results are consistent with SZ Lyn being a Population I star of normal abundance in an early stage of evolution off the main sequence.

I. INTRODUCTION

SZ Lyncis is an ultrashort-period, 0.12 day, pulsating variable. It has been classified at various times as a dwarf Cepheid, AI Velorum, or δ Scuti type. The more recent work seems to place it in a subset of δ Scuti type variables having large amplitudes.

What makes SZ Lyn exceptional is that it is in a binary system. This binary nature was first suggested by Barnes and Moffett (1975) as an explanation for the periodic ephemeris needed to account for the observed times of maximum. The radial-velocity observations of Bardin and Imbert (1981) confirmed this binary hypothesis.

In this paper, we report new photometric and radial-velocity observations of SZ Lyn. These new data, together with existing material in the literature, are used to determine improved orbital and pulsational properties for this interesting variable.

II. DATA

a) Photometry

The photometric observations were obtained over the period 1975 January 16 to 1979 March 4 (JD 2442428–2443936) with the 91 cm reflector at McDonald Observatory using the Purdue high-speed photometer described by Barnes, Evans, and Moffett (1978). During most of this period, the observations were made solely for the purpose of determination of times of maxima. However, over the period 1977 April 15 to 1979 March 4 (JD 2443248–2443936), SZ Lyn was differentially observed, on the *BVRI* system, with respect to the comparison stars listed in Table I. During this time period, observations were made over the complete pulsation cycle. These SZ Lyn data were reduced relative to each of the comparison stars, and the mean values adopted.

A total of 3760 observations in each of the four bandpasses were made and then condensed into the normal points given in Table II. Figures 1 and 2 show the light and color curves of these data, which will be used in a later paper to determine the radius of SZ Lyn by means of the visual surface-brightness method.

Table III gives the photoelectrically determined times of pulsation maxima used in the present analysis, including 13 new times of pulsation maxima measured in the present study. The five times of maxima for cycles, $E = 2919, 2977, 18\,459, 21\,073$, and $57\,701$, reported as suspect by Szeidl (1983) and Soliman *et al.* (1986) were deleted. In addition, we removed as suspect the observations of Joshi and Srivastava (1967) at $E = 2828$ and Jiang Shi-yang *et al.* (1985) at $E = 57\,036$.

b) Radial Velocities

The radial-velocity observations reported here were acquired over the period 1978 February to 1983 January with a photoelectric radial-velocity meter at the coude focus of the McDonald Observatory 2.1 m reflector. The instrument has been described in detail by Slovak *et al.* (1979) and Barnes *et al.* (1986). We used an entrance slit of 0.88 arcsec, which

TABLE I. Comparison stars for SZ Lyncis.

	V	(B - V)	(V - R)	(R - I)
BD +44°1714	9.685	0.624	0.540	0.346
BD+45°1544	9.475	0.431	0.409	0.259

TABLE II. *BVR*I normal points for SZ Lyncis.

ϕ	V	(B - V)	(V - R)	(R - I)	ϕ	V	(B - V)	(V - R)	(R - I)
0.00	9.151	0.210	0.192	0.091	0.45	9.566	0.336	0.323	0.185
0.01	9.152	0.207	0.189	0.097	0.55	9.634	0.337	0.339	0.197
0.02	9.152	0.207	0.189	0.084	0.56	9.631	0.339	0.338	0.196
0.03	9.158	0.211	0.194	0.080	0.57	9.635	0.337	0.333	0.204
0.04	9.170	0.212	0.200	0.079	0.58	9.645	0.341	0.349	0.194
0.05	9.176	0.217	0.206	0.079	0.59	9.652	0.337	0.344	0.179
0.06	9.187	0.218	0.199	0.103	0.60	9.657	0.345	0.346	0.201
0.07	9.197	0.221	0.205	0.111	0.61	9.658	0.350	0.346	0.195
0.08	9.207	0.229	0.208	0.121	0.62	9.660	0.351	0.345	0.208
0.09	9.210	0.231	0.204	0.109	0.63	9.666	0.354	0.351	0.214
0.10	9.231	0.226	0.206	0.093	0.64	9.670	0.352	0.344	0.210
0.14	9.270	0.244	0.237	0.157	0.65	9.661	0.335	0.341	0.192
0.15	9.286	0.248	0.233	0.145	0.66	9.666	0.343	0.347	0.179
0.16	9.297	0.251	0.233	0.152	0.67	9.666	0.351	0.344	0.199
0.17	9.311	0.256	0.244	0.170	0.68	9.673	0.340	0.355	0.192
0.18	9.325	0.259	0.248	0.176	0.69	9.667	0.339	0.347	0.197
0.19	9.333	0.265	0.251	0.177	0.70	9.664	0.357	0.338	0.199
0.20	9.354	0.263	0.268	0.170	0.71	9.669	0.345	0.344	0.218
0.21	9.359	0.263	0.256	0.147	0.72	9.672	0.339	0.356	0.209
0.22	9.372	0.273	0.264	0.162	0.77	9.643	0.349	0.320	0.172
0.23	9.378	0.278	0.264	0.151	0.78	9.632	0.345	0.316	0.176
0.24	9.400	0.281	0.275	0.143	0.79	9.626	0.341	0.319	0.169
0.25	9.402	0.281	0.271	0.139	0.80	9.610	0.343	0.309	0.180
0.26	9.415	0.286	0.270	0.156	0.81	9.607	0.328	0.316	0.177
0.27	9.423	0.286	0.282	0.149	0.82	9.591	0.326	0.308	0.174
0.28	9.434	0.290	0.277	0.158	0.83	9.577	0.325	0.306	0.167
0.29	9.444	0.295	0.289	0.168	0.84	9.555	0.329	0.296	0.161
0.30	9.457	0.292	0.293	0.151	0.85	9.539	0.319	0.285	0.160
0.31	9.475	0.291	0.300	0.148	0.86	9.510	0.299	0.283	0.157
0.32	9.483	0.303	0.297	0.157	0.87	9.482	0.290	0.274	0.151
0.33	9.494	0.300	0.305	0.152	0.88	9.454	0.280	0.264	0.146
0.34	9.498	0.303	0.301	0.170	0.89	9.424	0.279	0.258	0.134
0.35	9.502	0.306	0.302	0.171	0.90	9.389	0.265	0.251	0.127
0.36	9.501	0.314	0.302	0.170	0.91	9.337	0.257	0.229	0.129
0.37	9.509	0.316	0.309	0.168	0.92	9.299	0.245	0.225	0.098
0.38	9.521	0.315	0.318	0.187	0.93	9.266	0.228	0.212	0.101
0.39	9.522	0.319	0.311	0.177	0.94	9.234	0.229	0.205	0.104
0.40	9.533	0.322	0.315	0.169	0.95	9.215	0.221	0.204	0.102
0.41	9.539	0.321	0.321	0.173	0.96	9.188	0.217	0.200	0.091
0.42	9.548	0.326	0.322	0.176	0.97	9.171	0.213	0.189	0.095
0.43	9.553	0.335	0.323	0.177	0.98	9.160	0.211	0.191	0.086
0.44	9.562	0.333	0.324	0.183	0.99	9.152	0.209	0.185	0.088

projects to 34 km s^{-1} in the spectrum. We followed the reduction procedure described by Barnes *et al.* (1986). Using the observations of standard stars given in Barnes *et al.* (1986), we estimate the uncertainty of a single observation to be in the range of $\pm 4.0 \text{ km s}^{-1}$. The radial-velocity measurements for SZ Lyn are given in Table IV.

III. ANALYSES

All analyses of the data were done using the program GAUSSFIT (Jefferys 1987; Jefferys *et al.* 1987, 1988). This program allows one to perform a linear or nonlinear, iterative least-squares fit to the data. It correctly handles the case where more than one observation appears in each equation of condition (the errors-in-variables model) and includes a robust estimation feature that reduces the influence of outliers on the solution. GAUSSFIT is easy to use, even for com-

plex nonlinear problems, and was well suited to the task at hand.

a) Light-Travel-Time Orbit

It was first reported by van Genderen (1967) that a linear ephemeris did not accurately predict the times of pulsational maximum for SZ Lyn. Barnes and Moffett (1975) suggested that the nonlinear ephemeris for the times of maxima could be explained as due to orbital motion of SZ Lyn in a binary system. Deviations from a linear ephemeris were due to the light-travel time across the orbit.

Soliman *et al.* (1986) examined all available photoelectrically determined times of maxima and were able to fit these with the following relation:

$$T_{\max} = T_0 + P_0 E + \frac{\beta}{2} E^2 - A \cos 2\pi \left\{ \frac{P_0}{P_B} E - \psi_0 \right\}, \quad (1)$$

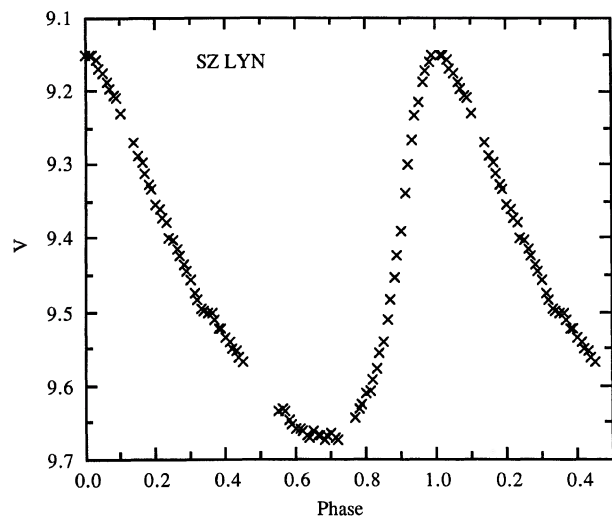


FIG. 1. V magnitude (normal points) versus pulsational phase for SZ Lyn-cis.

where E is the cycle number, T_0 is the initial epoch of maximum corresponding to $E = 0$, P_0 is the pulsation period, P_B is the period of the sinusoid, β is the secular change in the pulsation period per cycle, A is the semiamplitude of the sine curve, and ψ_0 is the phase shift. The above relation fits the observations quite well and shows no systematic deviations. Using Eq. (1), Soliman *et al.* (1986) found the pulsation period to be 0.120534896 days and the orbital period to be 1173.5 ± 2.0 days. Our observations provided an additional 13 times of maxima to the data set and the resulting fit, using Eq. (1), produced the same values of the fitted parameters as the Soliman *et al.* (1986) results. We want to emphasize that the form of Eq. (1) assumes a circular orbit and the resulting fit to the data is very good.

Irwin (1952) derived the following equation for the light-travel time across the orbit of a spectroscopic binary:

$$\tau = \frac{a \sin i}{2.59 \times 10^{10}} \left[\frac{1 - e^2}{1 + e \cos \nu} \sin(\nu + \omega) + e \sin \omega \right], \quad (2)$$

where τ is the light-travel time, a is the semimajor axis, e is the eccentricity, ν is the true anomaly, and ω is the longitude of periastron passage. Using the program GAUSSFIT, we fitted Eq. (2) to the $(O - C)$ values resulting from a linear ephemeris and found, without great surprise, that a circular orbit yielded the best fit. The fit was slightly improved by including the secular change in the pulsation period, but the orbit remained circular.

b) The Orbit from Radial Velocities

Bardin and Imbert (1981) measured 86 radial velocities of SZ Lyn using CORAVEL over the interval 1979 November 27–28 and 114 velocities over the interval 1980 March 24–25. These data allow one to construct accurate velocity curves for these two epochs. The above authors found the mean velocities for SZ Lyn at these epochs to be 41.11 km s^{-1} at HJD 2444206 and 43.36 km s^{-1} at HJD 2444324. We used these two radial-velocity curves as master curves.

The radial velocities given in Table IV allow us to determine the mean velocities for an additional seven epochs. Each of the master curves was shifted until good agreement with our velocities at each epoch was obtained. The mean velocity was determined with respect to each of the two master curves and then averaged. The mean velocities determined from our observations are given in Table V.

Bardin and Imbert (1984) gave additional mean velocities, determined from CORAVEL data, for SZ Lyn at 21 epochs. These CORAVEL data are of very high quality, with uncertainties of only $\pm 0.2 \text{ km s}^{-1}$. However, these observations only span a time window of 1454 days, or 1.23 orbital cycles. In order to expand the time baseline and improve the orbital-period determination, Bardin and Imbert (1984) included one earlier, but much less accurate, velocity due to Woolley and Aly (1966), which lengthened the time interval to 6944 days or 5.88 orbital cycles.

In fact, the mean velocity of Woolley and Aly (1966)

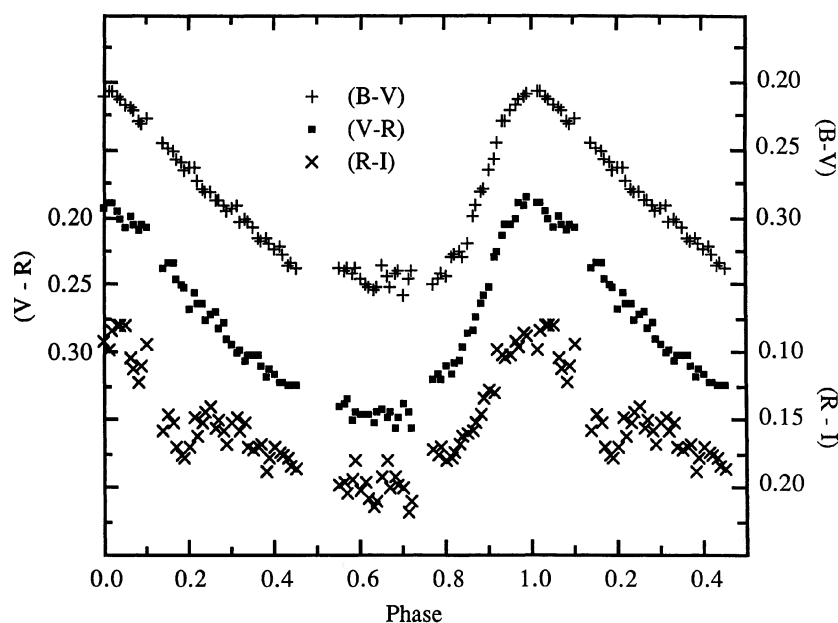


FIG. 2. Color curves (normal points) versus pulsational phase for SZ Lyn-cis.

TABLE III. Observed times of maximum light for SZ Lyncis.

Obs. Max HJD	E	(O - C)	ψ	Ref.	Obs. Max HJD	E	(O - C)	ψ	Ref.
2437367.4410	-6280	-2.37E-03	0.8609	Sc	2439052.5129	7700	-1.43E-03	0.2876	vG
2437368.2880	-6273	8.60E-04	0.8616	Sc	2439052.6355	7701	6.36E-04	0.2877	vG
2437368.4070	-6272	-6.78E-04	0.8617	Sc	2439054.5631	7717	-2.75E-04	0.2893	vG
2437642.0210	-4002	-9.12E-04	0.0934	Eg	2439054.6846	7718	6.93E-04	0.2894	vG
2437706.2645	-3469	-5.40E-04	0.1477	Sz	2439092.7729	8034	8.16E-04	0.3217	Bi
2437707.4695	-3459	-8.50E-04	0.1488	Sz	2439092.8931	8035	4.84E-04	0.3218	Bi
2437707.5908	-3458	-8.05E-05	0.1489	Sz	2439121.7004	8274	4.84E-04	0.3462	Bi
2437718.5587	-3367	-4.97E-04	0.1582	Sz	2439121.8208	8275	3.51E-04	0.3463	Bi
2437726.5141	-3301	-1.39E-04	0.1649	Sz	2439130.4981	8347	-7.17E-04	0.3536	vG
2437744.3535	-3153	6.83E-04	0.1800	Sz	2439145.4461	8471	1.18E-03	0.3663	vG
2437780.0310	-2857	1.01E-03	0.2102	HX	2439145.5642	8472	-1.25E-03	0.3664	vG
2437780.1520	-2856	1.47E-03	0.2103	HX	2439205.4706	8969	-3.75E-05	0.4171	vG
2437977.3430	-1220	2.01E-03	0.3773	HX	2439527.5441	11641	7.24E-04	0.6898	Sz
2437997.3510	-1054	1.46E-03	0.3942	HX	2439528.6288	11650	5.81E-04	0.6907	Sz
2438001.5680	-1019	-2.17E-04	0.3978	vG	2439531.4010	11673	4.04E-04	0.6930	JS
2438001.6890	-1018	2.49E-04	0.3979	vG	2439905.3070	14775	-7.90E-04	0.0096	Sz
2438003.3780	-1004	1.78E-03	0.3993	HX	2440243.3996	17580	3.93E-04	0.2959	Sz
2438021.5760	-853	-8.27E-04	0.4147	vG	2441035.4430	24151	-6.08E-04	0.9665	Sz
2438021.6990	-852	1.64E-03	0.4148	vG	2441312.4276	26449	6.66E-04	0.2010	Sz
2438032.3050	-764	6.46E-04	0.4238	HX	2441390.4113	27096	5.57E-04	0.2670	Sz
2438052.4333	-597	-2.72E-04	0.4408	Sz	2441678.8475	29489	-2.05E-04	0.5112	BM
2438052.5545	-596	3.94E-04	0.4409	Sz	2441679.5714	29495	4.81E-04	0.5118	Sz
2438053.2760	-590	-1.31E-03	0.4416	HX	2441679.6913	29496	-1.54E-04	0.5119	Sz
2438055.2050	-574	-8.63E-04	0.4432	HX	2441683.7909	29530	1.23E-03	0.5154	BM
2438055.3268	-573	4.03E-04	0.4433	Br	2441683.9091	29531	-1.10E-03	0.5155	BM
2438055.4463	-572	-6.32E-04	0.4434	Br	2441684.0312	29532	4.61E-04	0.5156	BM
2438055.5681	-571	6.34E-04	0.4435	Br	2442106.3947	33036	-4.57E-04	0.8732	Sz
2438057.3752	-556	-2.83E-04	0.4450	Br	2442129.5391	33228	6.79E-04	0.8928	KM
2438057.4964	-555	3.83E-04	0.4451	Br	2442136.2887	33284	1.76E-04	0.8985	KM
2438057.6172	-554	6.49E-04	0.4452	Br	2442136.4091	33285	3.80E-05	0.8986	KM
2438082.4464	-348	-2.86E-04	0.4663	Sz	2442136.5294	33286	-2.00E-04	0.8987	KM
2438114.3880	-83	-4.71E-04	0.4933	Br	2442162.4450	33501	-1.04E-04	0.9207	KM
2438118.3670	-50	8.64E-04	0.4967	vG	2442428.7054	35710	1.85E-03	0.1461	ME
2438118.4850	-49	-1.67E-03	0.4968	vG	2442428.8234	35711	-6.78E-04	0.1462	ME
2438124.3930	0	9.58E-05	0.5018	vG	2442439.7921	35802	-3.06E-04	0.1555	ME
2438140.4230	133	-1.13E-03	0.5153	Br	2442451.4834	35899	-5.18E-04	0.1654	Sz
2438439.3565	2613	-3.69E-04	0.7684	Sz	2442454.4971	35924	-9.46E-05	0.1679	Sz
2438439.4772	2614	-2.08E-04	0.7685	Sz	2442470.7663	36059	-2.59E-03	0.1817	ME
2438457.3173	2762	1.76E-04	0.7836	Sz	2442471.7330	36067	-1.36E-04	0.1825	ME
2438457.4376	2763	-6.22E-05	0.7837	Sz	2442472.6997	36075	2.32E-03	0.1833	ME
2438460.2100	2786	-5.06E-05	0.7861	JS	2442472.8173	36076	-6.15E-04	0.1834	ME
2438463.3420	2812	-2.06E-03	0.7887	JS	2442531.7560	36565	-1.68E-03	0.2333	Af
2438464.1890	2819	1.18E-03	0.7895	JS	2442532.7210	36573	-9.26E-04	0.2342	Af
2438466.2380	2836	1.02E-03	0.7912	JS	2442533.6852	36581	-9.78E-04	0.2350	Af
2438467.2000	2844	-1.29E-03	0.7920	JS	2442743.8944	38325	-6.60E-04	0.4130	ME
2438471.2970	2878	-2.60E-03	0.7955	JS	2442773.9089	38574	8.14E-04	0.4384	ME
2438701.8878	4791	5.36E-04	0.9907	Bi	2442774.0285	38575	-1.20E-04	0.4385	ME
2438708.6377	4847	5.08E-04	0.9964	vG	2442837.3091	39100	-3.49E-04	0.4920	Du
2438709.6014	4855	-6.63E-05	0.9972	vG	2442841.2875	39133	3.84E-04	0.4954	Du
2438712.6148	4880	-2.26E-05	0.9998	vG	2442871.4218	39383	7.76E-04	0.5209	KM
2438725.5117	4987	-2.62E-04	0.0107	vG	2442872.3861	39391	7.89E-04	0.5217	KM
2438752.5114	5211	-5.16E-05	0.0336	vG	2442874.3152	39407	1.32E-03	0.5234	KM
2438788.5505	5510	-2.03E-04	0.0641	vG	2442874.4352	39408	7.80E-04	0.5235	KM
2438809.4022	5683	-5.36E-04	0.0817	vG	2443232.4309	42378	-7.52E-04	0.8266	HW
2438830.4949	5858	-8.71E-04	0.0996	vG	2443248.7028	42513	-1.56E-03	0.8404	ME
2438834.2340	5889	1.76E-03	0.1028	JS	2443257.3835	42585	3.75E-04	0.8477	GE
2438844.4771	5974	-3.11E-04	0.1114	vG	2443258.4685	42594	5.30E-04	0.8486	GE
2438849.5396	6016	-1.25E-04	0.1157	vG	2443287.3985	42834	1.35E-03	0.8731	GE
2438850.3839	6023	4.57E-04	0.1164	vG	2443288.3600	42842	-1.45E-03	0.8739	GE
2438850.5034	6024	-5.74E-04	0.1165	vG	2443578.3668	45248	-7.05E-04	0.1195	Sz
2438871.3567	6197	8.37E-04	0.1342	vG	2443644.6584	45798	-1.24E-03	0.1756	ME

TABLE III. (continued)

Obs. Max HJD	E	(O - C)	ψ	Ref.	Obs. Max HJD	E	(O - C)	ψ	Ref.
2443879.5758	47747	-7.25E-04	0.3745	Sz	2445001.2758	57053	-9.01E-04	0.3242	JE
2443931.4059	48177	-1.49E-04	0.4184	Sz	2445002.1192	57060	-1.23E-03	0.3249	JE
2443936.7099	48221	3.47E-04	0.4229	ME	2445002.2398	57061	-1.16E-03	0.3250	JE
2444222.3817	50591	1.38E-03	0.6647	Ga	2445390.3636	60281	-1.13E-03	0.6536	BE
2444257.4574	50882	4.83E-04	0.6944	Ga	2445739.2014	63175	-8.29E-04	0.9490	JE
2444261.3156	50914	1.46E-03	0.6977	Ga	2445740.0449	63182	-1.08E-03	0.9497	JE
2444261.4359	50915	1.22E-03	0.6978	Ga	2445740.1672	63183	6.82E-04	0.9498	JE
2444262.2787	50922	2.52E-04	0.6985	Ga	2445740.2870	63184	-5.42E-05	0.9499	JE
2444269.2711	50980	1.43E-03	0.7044	Ga	2445758.3670	63334	-4.35E-04	0.9652	SE
2444578.2099	53543	1.31E-03	0.9660	JE	2445759.3310	63342	-7.20E-04	0.9660	SE
2444590.1415	53642	-9.21E-05	0.9761	JE	2445761.2612	63358	9.11E-04	0.9677	SE
2444591.3465	53652	-4.44E-04	0.9771	JE	2445761.3813	63359	4.76E-04	0.9678	SE
2444592.1907	53659	1.01E-05	0.9778	JE	2445767.0468	63406	8.10E-04	0.9726	JE
2444606.2899	53776	-3.38E-03	0.9898	JE	2446038.4852	65658	3.03E-04	0.2024	SE
2444607.1374	53783	3.81E-04	0.9905	JE	2446038.6061	65659	6.72E-04	0.2025	SE
2444607.2570	53784	-5.53E-04	0.9906	JE	2446039.4493	65666	1.53E-04	0.2032	SE
2444610.2709	53809	-1.92E-05	0.9931	JE	2446466.5001	69209	6.60E-06	0.5648	SE
2444611.1149	53816	2.39E-04	0.9939	JE	2446467.4653	69217	9.14E-04	0.5656	SE
2445001.1553	57052	-8.68E-04	0.3241	JE					

Sc = Schneller (1961); Eg = Eggen (1962); Sz = Szeidl (1983); HX = He Tian-jian and Xiong Da-run (1964); vG = van Genderen (1963, 1967); Br = Broglia (1963); JS = Joshi and Srivastava (1967); Bi = Binnendijk (1968); BM = Barnes and Moffett (1975); KM = Karetnikov and Medvedev (1977, 1978); ME = This paper; Af = Africano (1978); Du = Duerbeck (1976); HW = Hopp and Witzigmann (1979); GE = Garrido et al. (1979); Ga = Garbusov (1980); JE = Jiang Shi-yang et al. (1985); BE = Braune et al. (1983); SE = Soliman et al. (1986).

should not have been used at all. Their Fig. 13 shows large individual velocity residuals to a *cosine curve* fit of the pulsational velocities. It was this cosine curve that was used to determine the mean velocity. As can be seen in Fig. 1 of Bardin and Imbert (1981), the pulsational velocity curve is definitely not sinusoidal but rather sawtooth in shape. In addition, the total pulsational amplitude of Woolley and Aly (1966) is only 24 km s^{-1} , compared to 40 km s^{-1} for Bardin and Imbert (1981). Thus, it appears that it was purely fortuitousness that the mean velocity of Woolley and Aly (1966) fit the velocity curve of Bardin and Imbert (1984) so well. The orbital solution, based on the above data, of Bardin and Imbert (1984) indicated that SZ Lyn had an orbital period of 1181.5 ± 1.4 days with an eccentricity of 0.191 ± 0.008 .

Since the orbital period found by Bardin and Imbert (1984) did not agree with the orbital period determined from the light-travel-time analysis and the inclusion of the Woolley and Aly (1966) point is suspect, we decided to re-determine the spectroscopic orbit solution. Even if the velocities of Woolley and Aly (1966) were not suspect, an additional problem with spectroscopic period determinations was pointed out by Popper and McAlister (1987). They caution that a least-squares analysis that depends on a single observation separated by a long time interval from the bulk of the observations can lead to an erroneous value for the formal uncertainty associated with that period determination.

We used the following well-known equations:

$$V = \frac{a \sin i}{13750P\sqrt{1-e^2}} [\cos(v+\omega) + e \cos \omega] + \gamma, \quad (3)$$

$$\tan \frac{1}{2}E = \sqrt{\frac{1-e}{1+e}} \tan \frac{1}{2}v, \quad (4)$$

$$\frac{t-T}{P} = \frac{M}{2\pi} = \frac{E - e \sin E}{2\pi}, \quad (5)$$

where most of the terms have been previously defined, except that E is the eccentric anomaly, not the cycle number, M is the mean anomaly, T is the time of periastron passage, and V is the observed radial velocity at time t . The above equations were fitted *only* to the CORAVEL data using GAUSSFIT. The solution gave an orbital period of 1186.9 ± 6.5 days with an eccentricity of 0.186 ± 0.006 . In view of the problem with the velocity of Woolley and Aly (1966) and the warning expressed by Popper and McAlister (1987), we feel that the true uncertainty in the orbital period determined from the spectroscopic data is closer to ± 5 –6 days than the quoted value of ± 1.4 days given by Bardin and Imbert (1984).

The seven mean velocities reported in Table V are consistent with the CORAVEL results but they do not improve the solution since our velocities were obtained within the same time frame as the CORAVEL data and have larger uncertainties.

IV. THE PUZZLE AND SOLUTION

The light-travel-time analysis is based on 159 photoelectric measurements (see Table III) of the times of maxima covering 9100 days or approximately 7.7 orbital periods. With these data one would expect to find a rather well-determined orbital period. The resulting (O-C)'s using the ephemeris given by Eq. (1) are small and show no systematic deviations from a mean zero as a function of orbital phase. The fact that the fit to Eq. (1), which assumes a circular orbit, is very good clearly seems to indicate that the orbit is circular. Using the light-travel time given by Eq. (2) and fitting the residual (O-C)'s, after correction for secular period change and a linear ephemeris, one still finds that a circular orbit is indicated.

The orbital solution based on the radial-velocity data clearly indicates an eccentricity that is nonzero. The high

TABLE IV. Radial velocities of SZ Lyncis.

HJD	V_r km/s	UT Date	HJD	V_r km/s	UT Date
2443563.812	15.1	1978 Feb 24	2444627.834	50.9	1981 Jan 23
2443618.640	8.1	1978 Apr 20	2444627.840	50.2	1981 Jan 23
2443619.696	32.2	1978 Apr 21	2444627.846	52.1	1981 Jan 23
2443621.618	35.3	1978 Apr 23	2444627.856	34.4	1981 Jan 23
2443820.913	16.8	1978 Nov 08	2444627.862	24.8	1981 Jan 23
2443820.928	21.9	1978 Nov 08	2444627.867	16.3	1981 Jan 23
2443821.915	44.9	1978 Nov 09	2444627.873	13.4	1981 Jan 23
2443821.925	45.3	1978 Nov 09	2444627.879	24.5	1981 Jan 23
2443821.941	39.4	1978 Nov 09	2444627.884	11.1	1981 Jan 23
2443821.952	18.7	1978 Nov 09	2444627.890	24.7	1981 Jan 23
2443825.957	12.6	1978 Nov 13	2444627.900	16.3	1981 Jan 23
2443825.968	11.5	1978 Nov 13	2444627.906	28.9	1981 Jan 23
2443882.872	6.6	1979 Jan 09	2444627.912	34.2	1981 Jan 23
2443882.879	15.8	1979 Jan 09	2444627.917	26.0	1981 Jan 23
2443882.888	36.2	1979 Jan 09	2444947.969	39.7	1981 Dec 09
2443882.896	42.1	1979 Jan 09	2444947.974	37.6	1981 Dec 09
2443882.904	35.5	1979 Jan 09	2444947.979	37.5	1981 Dec 09
2443882.925	51.3	1979 Jan 09	2444947.983	30.2	1981 Dec 09
2443885.828	15.0	1979 Jan 12	2444947.988	21.7	1981 Dec 09
2443885.845	12.9	1979 Jan 12	2444947.993	10.8	1981 Dec 09
2443885.856	3.0	1979 Jan 12	2444948.944	39.4	1981 Dec 10
2443940.712	18.8	1979 Mar 08	2444948.949	36.7	1981 Dec 10
2443940.726	18.7	1979 Mar 08	2444948.954	23.2	1981 Dec 10
2443940.788	46.6	1979 Mar 08	2444948.959	14.6	1981 Dec 10
2443940.801	8.7	1979 Mar 08	2444949.950	10.3	1981 Dec 11
2443941.689	17.7	1979 Mar 09	2444949.955	16.2	1981 Dec 11
2443941.707	34.0	1979 Mar 09	2444949.961	16.0	1981 Dec 11
2443941.718	36.8	1979 Mar 09	2444949.965	18.8	1981 Dec 11
2443941.729	36.1	1979 Mar 09	2445339.890	33.6	1983 Jan 05
2443941.798	23.2	1979 Mar 09	2445339.922	49.7	1983 Jan 05
2443941.807	20.0	1979 Mar 09	2445339.930	56.2	1983 Jan 05
2444627.811	41.4	1981 Jan 23	2445339.937	58.6	1983 Jan 05
2444627.817	39.5	1981 Jan 23	2445340.913	57.8	1983 Jan 06
2444627.823	52.6	1981 Jan 23	2445340.926	31.5	1983 Jan 06
2444627.829	46.7	1981 Jan 23			

quality of the CORAVEL data together with the reanalysis of the spectroscopic solution leads one to place high confidence in the spectroscopic determination.

Soliman *et al.* (1986) were the first to point out the above puzzle and speculated that the spectroscopic period might be in error due to the isolated point argument as proposed by Popper and McAlister (1987). Our analysis shows that the

spectroscopic and photometric orbital periods may be in much better agreement if a larger uncertainty, in the range of ± 5 –6 days, is assigned to the spectroscopic results of Bardin and Imbert (1984). Even if the period difference is reduced or eliminated, the eccentricity discrepancy remains. Resolution of the above puzzle is simple but, as is often the case, simple solutions are not always immediately obvious.

The light-travel time across a binary orbit is correctly given by Eq. (2), but the contribution to the $(O - C)$'s due to light travel time is made up of *two* components. One component is the periodic component, as expressed by Eq. (2), due to the orbital motion of SZ Lyn about the center of mass of the binary system. The other component, which has not been considered, is the contribution to the observed light-travel time caused by the motion of the binary's center of mass with respect to the Sun, i.e., the γ velocity of the system.

If one allows for the orbital motion of SZ Lyn about the center of mass, the motion of the center of mass of the binary system with respect to the Sun, the pulsation period, and the secular change in the pulsation period, the following ephemeris for the times of pulsation maximum results:

TABLE V. Mean velocities.

HJD	V_o km s ⁻¹
2443620.13	26.2
2443821.43	28.0
2443884.36	30.1
2443941.26	33.6
2444627.86	33.1
2444948.97	26.0
2445340.41	38.9

TABLE VI. Orbital parameters.

Source	P(orbital)	T	e	ω	$a_1 \sin i$
	days	HJD			$\times 10^6$ km
Bardin & Imbert(1984)	1181.5 ± 1.4	2445804.9 ± 7.9	0.191 ± 0.008	100°.6 $\pm 2^\circ.7$	152.4 ± 1.6
CORAVEL This Paper	1186.9 ± 6.5	2445810.5 ± 14.3	0.186 ± 0.006	101°.1 $\pm 3^\circ.4$	152.3 ± 1.0
Photometry This Paper	1176.6 ± 1.9	2445799.1 ± 48.0	0.18 ± 0.04	106°.6 $\pm 14^\circ.7$	147.2 ± 2.7

$$T_{\max} = T_0 + EP_0 \left[1 + \frac{\gamma}{c} \right] + \frac{\beta}{2} E^2$$

$$+ \frac{a \sin i}{2.59 \times 10^{10}} \left[\frac{1 - e^2}{1 + e \cos \nu} \sin(\nu + \omega) + e \sin \omega \right], \quad (6)$$

where all the terms are as previously defined, with the addition that γ is the gamma velocity of the binary system and c is the speed of light. The term $EP_0\gamma/c$ corrects for the motion of the center of mass of the system relative to the Sun. In Table VI we show a comparison of the spectroscopic and photometric results using the ephemeris given by Eq. (6). Once the photometric data are corrected for the light-travel time due to the motion of the center of mass with respect to the Sun, the solution indicates an eccentric orbit in good agreement with the spectroscopic results.

V. RESULTS

In order to determine the best values for the pulsational and orbital parameters, we made use of both the photometric and spectroscopic data in a combined iterative solution. We started the iterative solution by adopting the values of T , e , ω , γ , and $a_1 \sin i$ as given by Bardin and Imbert (1984), then used GAUSSFIT to solve Eqs. (4)–(6) for the remaining parameters. Using these results, improved values for the spectroscopic parameters were determined by a GAUSSFIT solution to Eqs. (3)–(5). This iterative procedure was continued until convergence was achieved. These final orbital and pulsational parameters are given in Table VII.

The major differences between our results and previous work are: the pulsation period of SZ Lyn is found to be shorter and the orbital period is now consistent with the spectroscopic results. In Fig. 3 we show the residual ($O - C$)'s after applying a linear ephemeris, corrected for secular period change and the motion of the center of mass with respect to the Sun. Also shown, as the solid curve, is the theoretical light-travel time, as given by Eq. (2), expected for an orbit with the parameters given in Table VII.

In Fig. 4 we show the final ($O - C$)'s determined from the ephemeris given by Eq. (6) using the values from Table VII. As this figure illustrates, the final ephemeris, Eq. (6), does a good job in predicting the observations over a long time base. The values of the final ($O - C$)'s are given in column 3 of Table III along with the corresponding orbital phase given in column 4. The orbital phases listed in Table III were calculated with respect to the time of periastron passage T given in Table VII. These are different from the

orbital phases given in the earlier papers by Barnes and Moffett (1975), Soliman *et al.* (1986), and Szeidl (1983), which were calculated with respect to T_0 , the time of the initial epoch of maximum corresponding to E (cycle number) = 0.

TABLE VII. Final pulsation and orbital parameters.

Parameter	Value
P_0 (pulsation)	0.120521154 days ± 0.000000012
T_0 ($E = 0$)	2438124.3987 HJD ± 0.0001
β	$+3.0 \times 10^{-12}$ days cycle $^{-1}$ $\pm 4 \times 10^{-12}$
P(orbital)	1181.1 days ± 8
e	0.188 ± 0.005
ω	99°.0 $\pm 2^\circ.3$
T	2445799.5 HJD ± 6.6
$a_1 \sin i$	152.3×10^6 km $\pm 8 \times 10^6$
γ or V_0	34.17 km s^{-1} ± 0.4
K	9.55 km s^{-1} ± 0.5
$f(M)$	$0.101 M_\odot$ ± 0.002

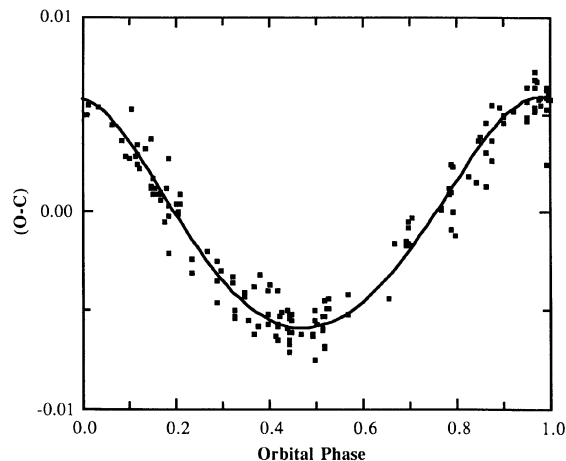


FIG. 3. Residuals in the time of maximum light, determined from a linear ephemeris corrected for motion of the center of mass of the binary with respect to the Sun and for secular change in the pulsation period of SZ Lyn. The solid curve is the theoretical light-travel-time curve for a binary with the orbital parameters as given in Table VII versus orbital phase.

VI. DISCUSSION

Using the mass function together with the known properties of SZ Lyn, we can place a few restrictions on the mass of the unseen companion. The lack of spectroscopic detection of the companion suggests that it must be later than spectral type F2 V, as discussed by Barnes and Moffett (1975) and Bardin and Imbert (1984). In the following discussion, we will assume that the companion is a normal star. In terms of mass, this limitation implies an upper limit for the companion of about $1.6 M_{\odot}$. It is possible that the companion could be a neutron star or black hole, but since there are no compelling reasons to adopt this assumption, we will confine ourselves to normal stars.

McNamara and Feltz (1978) used *uvby* β data to determine a pulsation mass of $1.53 M_{\odot}$ for SZ Lyn and a value of $1.61 M_{\odot}$ based upon fitting their data to theoretical models. Fernley *et al.* (1984) used *BVJHK* photometry and radial velocities to perform a Baade-Wesselink analysis, which yielded a pulsation mass of $1.57 M_{\odot}$ for the variable. In considering all the uncertainties, they place the mass of SZ Lyn in the range of 0.91 – $1.74 M_{\odot}$. We will adopt the unweighted mean of the above three estimates, $1.57 M_{\odot}$, along with the uncertainty as stated by Fernley *et al.* (1984).

In this binary system, the fact that eclipses have not been observed does not place any significant limit on the allowable orbital inclinations. Lack of eclipses only implies that the inclination is less than 89° .

The change in the pulsation period determined in this paper can be compared to the evolutionary tracks, based on the models of Hallgren (1966), as given in Percy *et al.* (1980). The secular pulsation-period change given in Table VII is consistent with the evolutionary models for a Population I star with normal metal abundances just evolving away from the main sequence. One must exercise caution in using the period change result since there are significant uncertainties in both the observations and theory. The secular period change seems to favor a high mass for SZ Lyn in the range of 1.5 – $2.0 M_{\odot}$.

Figure 5 shows the various possibilities for the masses of the two components of this binary along with the various

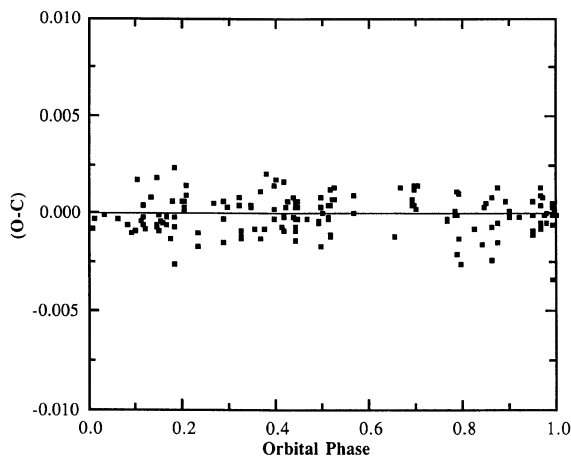


FIG. 4. Residuals in the time of maximum light, determined using the complete ephemeris, Eq. (6), versus orbital phase.

restrictions discussed above. The upper boundary is defined by the mass of an F2 V star, and the lower boundary corresponds to an orbital inclination of 90° . The middle vertical line is the mean of the mass determinations for SZ Lyn, and the two other vertical lines reflect the estimated uncertainties on the mass of the variable. The two thatched regions in the figure encompass the range of probable solutions to the mass function. The smaller thatched area represents the more probable solutions. We believe the most likely mass for the companion to be in the range 0.7 – $1.6 M_{\odot}$ and thus the spectral type to be F2 V–K3 V.

VII. CONCLUSIONS

We have derived an ephemeris based upon an eccentric orbit which predicts the observed times of light maxima to good precision and is in agreement with the spectroscopic results. The secular change in the pulsation period of SZ Lyn

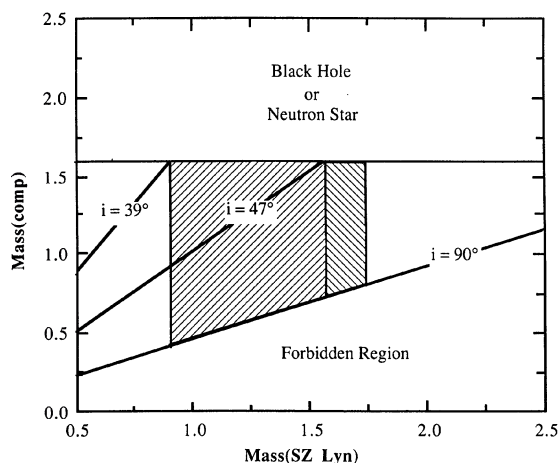


FIG. 5. Solutions of the mass function for various orbital inclinations. The upper boundary corresponds to the mass of an F2 V star, and the lower boundary is defined by an inclination of 90° . The middle vertical line represents the mean of three mass determinations for SZ Lyn, and the two other vertical lines represent the estimated limits on the mass of the variable. The two thatched regions represent the range of probable solution of the mass function. The smaller thatched area is the more probable range of solutions.

is consistent with a Population I star of normal abundance just leaving the main sequence.

Based on the known properties of SZ Lyn, we can state that the orbital inclination is probably greater than 39° and the unseen companion is in the range of F2 V–M2 V. If our interpretation of the secular period change is correct, we can tighten the above limits. We feel that it is more probable that the orbital inclination is greater than 47° and the companion star is in the range of F2 V to K3 V. The possibility that the companion is a white dwarf, neutron star, or black hole cannot be excluded, but there are no reasons to suspect that is the case.

This work formed part of a much larger program of radial velocity and photometric observations conducted at McDonald Observatory. We wish to express our appreciation to the dedicated efforts of many night assistants who operated the telescope for us. Financial assistance for this research is gratefully acknowledged from NSF grants AST 76-20841 (T.J.M.), AST 79-19913 (T.J.M.), AST 76-21316 (T.G.B.), and AST 79-19826 (T.G.B.). GAUSSFIT was developed under NASA contract NAS8-32906 (W.H.J.) for Space Telescope Astrometry. Part of this work was supported by a grant from the Vanderbilt University Research Council (F.C.F.).

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